

can strain their budgets. Certification costs can vary widely depending on the product, region, and standards involved. These costs can include testing fees, consultant fees, and potential redesign expenses if the initial design doesn't pass certification.

Delays in time-to-market. Certification processes can be time-consuming. Delays in obtaining certifications can lead to postponed product launches and missed market opportunities. This can be especially challenging for startups trying to establish themselves in a competitive market.

Unpredictable costs. The actual certification costs might be hard to predict accurately. Testing failures or the need for design changes to meet standards can lead to unexpected expenses.

So, is there no way to avoid this?

Well, yes, there is a way. First of all, if you are a hobbyist, make five devices from a single design without requiring any testing as per section 15.23 of 47 CFR Ch.1: However, there are several key points you need to consider:

Personal use. The exemption is generally intended for personal use and not for commercial purposes. It's not intended to allow individuals to manufacture and sell devices as a business.

Limited distribution. The exemption often includes restrictions on distribution. In many cases, the devices exempted under this rule are not allowed to be sold or distributed in significant quantities.

Modifications. The devices must be constructed in accordance with the regulations, and any modifica-

Non-IAF ISO 9001:2015 Certification	IAF Approved ISO 9001:2015 Certification
Rs. 1500/-	Rs. 3999/-
Get Certificate in 2 Working Days Used for Marketing & Branding Purposes only	Get Certificate in 2 Working Days Valid in Government and Private Tenders by all Nations
Call Now for More Details +91 8591017450	Call Now for More Details +91 8591017450
Get Quotation >	Get Quotation >

Fig. 1: ISO price chart (Source: <https://isoconsultancy.in>)

BIS CERTIFICATION COST FOR ISI MARK	
Type of Fees/Charges	Amount (₹)
Application fee	1,000
Audit charges	7,000
Sample test fee	As applicable
License fee	1,000

(Source: <http://surl.li/lbmzg>, <https://alephindia.in>)

tions made by the hobbyist should not exceed the limits set forth by the regulations.

Interference. The devices built under this exemption must not cause harmful interference to other authorised radio services.

Labelling and identification. Devices constructed under this exemption might still need to comply with labelling and identification requirements to indicate that they are not FCC-certified.

It's important to note that while this exemption might allow hobbyists to build a small number of devices without certification, if you are planning to sell devices in any significant quantity or for commercial purposes, you are likely to be subject to FCC certification requirements. Additionally, regulations can change, so it's always a good idea to consult the latest version of the FCC rules or seek legal advice to ensure

you are in compliance with the regulations.

For India BIS, ISI, and ISO certification, you can get these at an affordable cost easily.

So, if you are an engineer who wants to share and design for manufacturing as a hobbyist, then you no longer need certification to sell your design. But if you are a startup and launching the product design in the market at a large scale, you can proceed further.

Ways to avoid FCC certification costs

If you want to avoid the high FCC certification costs, it starts at the component selection stage. Yes, before proceeding with the design, you need to select the right components for your design. Luckily and unknowingly, I selected the correct components in my design that gave a huge cost relief and high ROI on each product sale. Here is how:

The FCC provides a certification option known as Modular Certification, which enables designers to utilise pre-certified wireless modules in their designs. This means you can purchase modules that have already been FCC-certified and integrate them into your own product design, eliminating the certification cost.

There are companies that launch these modules along with the chips. They go through all the documentation, testing, and certification processes for you, helping you. For example, Fig. 2 shows a Wi-Fi BLE module. If you use this module to design any development board or IoT device, and don't change much in the RF design, like adding an external antenna, and